Lab 1 Report

System Identification

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1 Introduction

The aim of this laboratory was to identify a workable dynamic model of the suspension simulator using two complementary experiments: a step-response test and a frequency-response test.

The laboratory system is a simulated spring–mass–damper system whose parameters are unique to the student. The lab brief specifically requires the time-domain response to be used to estimate the gain, pole type and dominant pole location, and then requires sinusoidal tests to confirm those estimates in the frequency domain [1].

The measured system was found to be a lightly overshooting but relatively well-damped second-order system. The step test was first used to estimate the steady-state gain, damping ratio, natural frequency and complex pole pair.

Nine sinusoidal tests were then performed between 0.500 rad/s and 8.000 rad/s, with extra points placed around the frequency predicted by the step test.

Finally, the independently estimated time- and frequency-domain models were combined and validated against both experimental datasets.

All raw measurements, processed results, MATLAB analysis scripts, generated figures and the report source are archived in the public project repository [3]. Direct links to the main reproducibility files are provided in Section A.

2 System Modelling

2.1 Physical Model

The simulated suspension is described in the lab brief by

$$m\ddot{x}(t) = -b\dot{x}(t) - kx(t) + F(t), \tag{1}$$

where $x(t)$ is displacement, b is the damping coefficient, k is the spring coefficient, and $F(t)$ is the applied force.

The lab specifies that the parameters are scaled to the mass, so $m = 1$, and that the applied force is directly proportional to the simulator voltage input [1].

Let the proportionality constant between simulator input $u(t)$ and force be c, such that

$$F(t) = cu(t). \tag{2}$$

Equation (1) therefore becomes

$$\ddot{x}(t) + b\dot{x}(t) + kx(t) = cu(t). \tag{3}$$

Taking the Laplace transform with zero initial conditions gives

$$G(s) = \frac{X(s)}{U(s)} = \frac{c}{s^2 + bs + k}. \tag{4}$$

For identification, the same transfer function can be written in the standard second-order form

$$G(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}, \tag{5}$$

where K is the DC gain, ζ is the damping ratio, and ω_n is the natural frequency.

Comparing Equations (4) and (5) gives

$$b = 2\zeta\omega_n, \qquad k = \omega_n^2, \qquad c = K\omega_n^2. \tag{6}$$

2.2 Experimental Method

The suspension simulator was operated in “Laboratory Testing” mode.

The manual states that the logged input and output displacement are sampled every 100.000 ms [2].

The recorded CSV files were analysed as follows:

1. For the step test, the step instant was shifted to $t = 0$ and the initial displacement was subtracted from the output. The actual recorded input step was $\Delta u = 4$, so all gain calculations were normalised by this known step size rather than assuming a unit step.
2. For the frequency test, sinusoidal tests were performed at nine angular frequencies from 0.5 to 8 rad/s, with additional measurements around the predicted corner region.
   The measured input was approximately a sinusoid of amplitude 2 about a DC offset of 4. The offset does not affect the sinusoidal gain calculation for an LTI model; only the AC amplitudes were used.
3. At each test frequency, the first four cycles were discarded to reduce transient contamination. A sine/cosine least-squares fit was then used to determine the input and output amplitudes and phase angles.
4. The resulting model was validated against both the measured step response and the measured frequency response.

The complete MATLAB implementations used to reproduce these calculations are available in the repository [3]; direct links are listed in Table 11.

The two experiments and the information extracted from each are summarised in Table 1.

Table 1: Summary of experimental tests.

Test	Input	Purpose
Step response	$0 \to 4$	Estimate the DC gain, damping ratio, natural frequency and complex pole pair.
Frequency response	0.5–8 rad/s	Confirm the gain and natural-frequency region using measured magnitude and phase.
Model validation	Measured datasets	Compare the proposed transfer function with the experimental time- and frequency-domain responses.

3 System Identification

3.1 Step Response Test

Figure 1 shows the recorded simulator input.

In the raw log, the input changes from 0 to 4 at approximately

$$t = 16.300\,\mathrm{s}.$$

For analysis, this instant was shifted to $t = 0$.

Because the step size is known, the use of a 4-unit step does not change the identification method; the steady-state output change is simply divided by four.

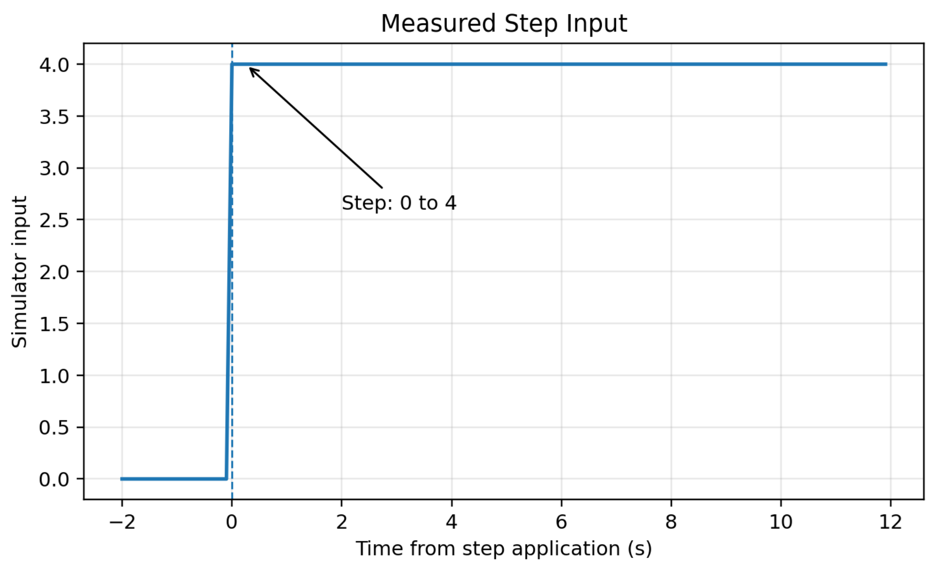

Figure 1: Measured step input. The recorded input changes from 0 to 4 at $t = 0$.

The initial displacement was

$$y_0 = 5.000$$

and the final displacement was

$$y_{ss} = 7.552.$$

The steady-state output change was therefore

$$\Delta y_{ss} = 7.552 - 5.000 = 2.552 \tag{7}$$

The DC gain is therefore

$$\boldsymbol{K_{\text{step}} = \frac{\Delta y_{ss}}{\Delta u} = \frac{2.552}{4} = 0.638} \tag{8}$$

The principal measured values used in the step-response calculations are collected in Table 2.

Table 2: Measured quantities from the step-response test.

Quantity	**Measured Value**
Raw step time	16.3 s
Input change, Δu	$0 \rightarrow 4$
Initial displacement, y_0	5.000
Final displacement, y_{ss}	7.552
Steady-state displacement change, Δy_{ss}	2.552
First peak displacement, y_{peak}	7.628
Peak time relative to the step, t_p	2.6 s

The zero-adjusted response is shown in Figure 2.

A small but clear overshoot is present, confirming that the poles are a complex-conjugate pair and that

$$0 < \zeta < 1.$$

In the raw log the first peak occurs at approximately

$$t = 18.900\,\text{s}.$$

Relative to the step at

$$t = 16.300\,\text{s},$$

the peak time is

$$t_p = 18.9 - 16.3 = 2.600\,\text{s} \tag{9}$$

The corresponding peak displacement is

$$y_{peak} = 7.628.$$

The zero-adjusted peak displacement is therefore

$$2.628.$$

Hence,

$$\boldsymbol{M_p = \frac{2.628 - 2.552}{2.552} = 0.0298 = 2.98\%} \tag{10}$$

The percentage overshoot is therefore approximately

$$\boxed{\boldsymbol{\%OS \approx 2.98\%}}$$

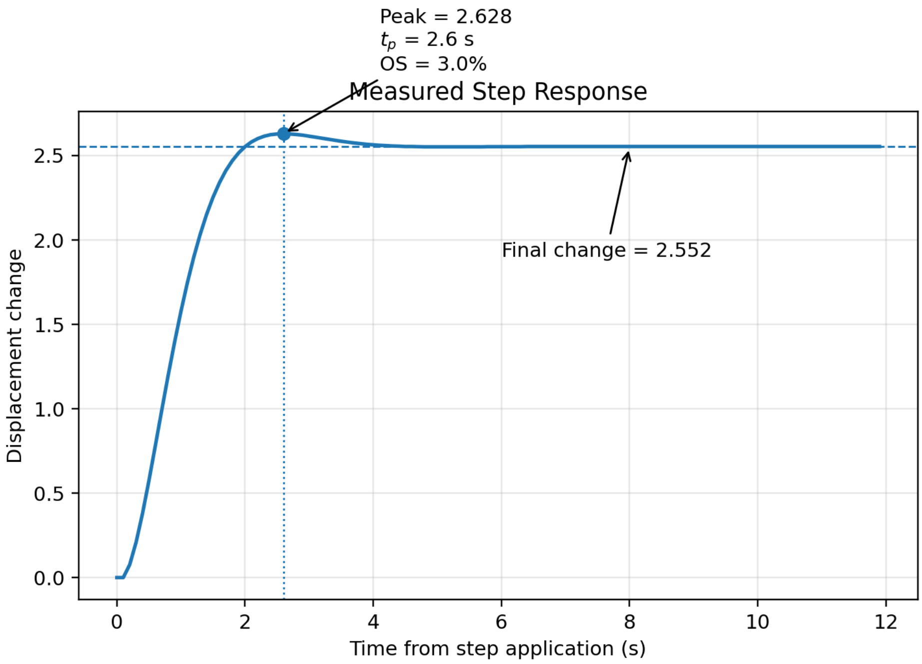

Figure 2: Measured zero-adjusted step response. The final displacement change, peak value, peak time and overshoot are annotated.

For an underdamped second-order system,

$$M_p = e^{-\zeta\pi/\sqrt{1-\zeta^2}} \tag{11}$$

The damping ratio can therefore be calculated directly as

$$\zeta_{\text{step}} = \frac{-\ln(M_p)}{\sqrt{\pi^2 + [\ln(M_p)]^2}} = \mathbf{0.745} \tag{12}$$

The damped natural frequency follows from the peak time:

$$\omega_d = \frac{\pi}{t_p} = \frac{\pi}{2.6} = \mathbf{1.208}\,\text{rad/s} \tag{13}$$

Thus,

$$\omega_{n,\text{step}} = \frac{\omega_d}{\sqrt{1-\zeta^2}} = \mathbf{1.813}\,\text{rad/s} \tag{14}$$

The dominant complex pole pair is therefore

$$\boldsymbol{s} = -\zeta\omega_n \pm j\omega_d = \mathbf{-1.351 \pm j1.208} \tag{15}$$

The measured 2% settling time is approximately

$$T_s \approx \mathbf{3.2}\,\text{s} \tag{16}$$

The standard estimate is

$$\frac{4}{T_s} \approx \mathbf{1.25}\,\text{s}^{-1} \tag{17}$$

which is reasonably close to the calculated pole real part,

$$\zeta\omega_n \approx \mathbf{1.35}\,\text{s}^{-1} \tag{18}$$

Substitution into Equation (5) gives the step-test model

$$\boxed{\boldsymbol{G_{\text{step}}(s) = \frac{2.097}{s^2 + 2.703s + 3.287}}} \tag{19}$$

The corresponding parameter estimates are

$$\boldsymbol{b_{\text{step}} = 2.703}, \qquad \boldsymbol{k_{\text{step}} = 3.287} \tag{20}$$

Table 3 summarises the complete parameter set obtained from Stage 1.

Table 3: Parameters identified from the step response.

Parameter	**Symbol**	**Estimate**
DC gain	K_{step}	0.638
Percentage overshoot	$\%OS$	2.98%
Damping ratio	ζ_{step}	0.745
Damped natural frequency	ω_d	1.208 rad/s
Natural frequency	$\omega_{n,\text{step}}$	1.813 rad/s
Dominant poles	$s_{1,2}$	$-1.351 \pm j1.208$
Damping coefficient	b_{step}	2.703
Spring coefficient	k_{step}	3.287

The important frequency predicted from the step response is therefore around

$$\omega_n \approx \mathbf{1.8}\,\text{rad/s} \tag{21}$$

The relative -3 dB point of this model is approximately

$$\omega_{-3\text{dB}} \approx \mathbf{1.71}\,\text{rad/s} \tag{22}$$

This estimate motivated the denser frequency sampling between

$$\mathbf{1.0} \le \omega \le \mathbf{2.2}\,\text{rad/s} \tag{23}$$

3.2 Frequency Response Test

The chosen frequencies span well below, around and well above the natural-frequency estimate obtained from the step test.

In particular, the points at

$$1.4, \quad 1.8, \quad 2.2 \text{ rad/s}$$

provide additional resolution around the expected corner region, while

$$3, \quad 5, \quad 8 \text{ rad/s}$$

reveal the high-frequency roll-off.

The choice of test points is summarised in Table 4.

Table 4: Rationale for the selected sinusoidal test frequencies.

Frequency band	**Test points (rad/s)**	**Purpose**
Low frequency	0.5, 0.8	Estimate the low-frequency/DC gain before the main roll-off
Transition region	1.0, 1.4	Observe the start of the magnitude decrease and increasing phase lag
Predicted corner region	1.8, 2.2	Resolve the region around the step-test estimate $\omega_n \approx 1.8$ rad/s
High frequency	3, 5, 8	Confirm the second-order roll-off and increasing phase lag

For each frequency, the sinusoidal input and output were represented as

$$q(t) = q_0 + a\sin(\omega t) + b\cos(\omega t) \tag{24}$$

The sinusoidal amplitude and phase were then calculated as

$$A_q = \sqrt{a^2 + b^2}, \qquad \phi_q = \tan^{-1}\left(\frac{b}{a}\right) \tag{25}$$

The frequency-response magnitude was calculated from

$$|G(j\omega)| = \frac{A_{\text{out}}}{A_{\text{in}}} \tag{26}$$

and in decibels,

$$G_{\mathrm{dB}} = 20\log_{10}|G(j\omega)| \tag{27}$$

The phase lag was obtained from

$$\phi = \phi_{\mathrm{out}} - \phi_{\mathrm{in}} \tag{28}$$

The measured magnitude and aligned phase values are listed in Table 5.

The raw phase measurements contained an additional frequency-proportional lag very close to

$$\omega(0.1 \text{ s}).$$

Since the manual specifies a 100.000 ms logging interval [2], this was treated as a one-sample alignment effect rather than part of the continuous-time plant.

The plotted phase values therefore use the correction

$$\phi_{\mathrm{aligned}} = \phi_{\mathrm{raw}} + \omega(0.1)\frac{180}{\pi} \tag{29}$$

This correction makes the measured phase consistent with the simultaneously measured magnitude and the step-response dynamics.

Table 5: Measured frequency-response data.

ω **(rad/s)**	$\lvert G(j\omega)\rvert$	**Magnitude (dB)**	**Aligned phase (deg)**
0.5	0.631	-4.00	-23.5
0.8	0.615	-4.22	-38.4
1.0	0.595	-4.51	-48.7
1.4	0.530	-5.52	-69.3
1.8	0.441	-7.10	-87.9
2.2	0.353	-9.05	-103.2
3.0	0.221	-13.10	-124.2
5.0	0.0875	-21.16	-148.0
8.0	0.0357	-28.96	-161.3

Figures 3 and 4 show the measured frequency-response points together with the final model introduced later in Section 3.3.

The magnitude falls monotonically rather than showing a resonance peak.

This is expected because a standard second-order low-pass system has a resonance peak only when

$$\zeta < \frac{1}{\sqrt{2}} \approx \mathbf{0.707} \tag{30}$$

The step response already indicated a damping ratio close to

$$\zeta \approx \mathbf{0.745} \tag{31}$$

Thus, the lack of a resonance peak is consistent with, rather than contradictory to, the small time-domain overshoot.

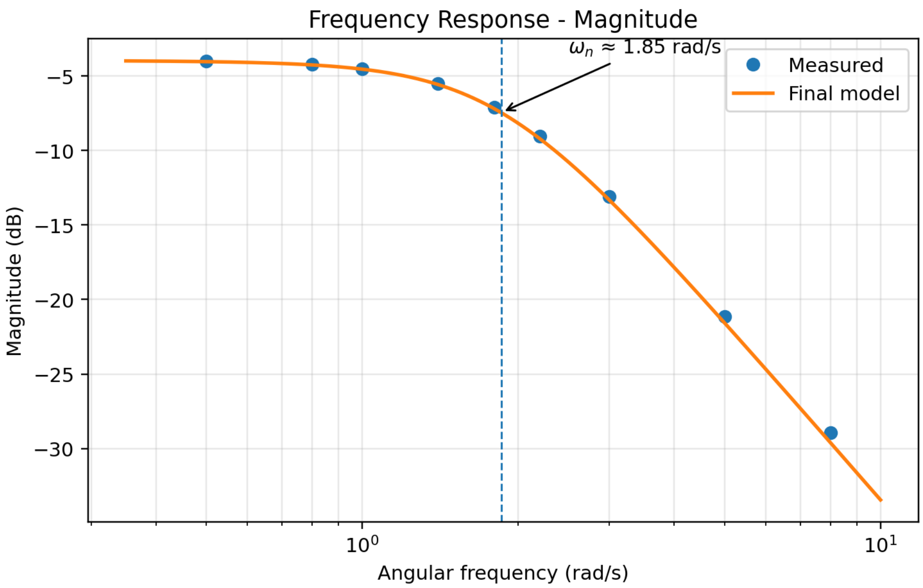

Figure 3: Measured frequency-response magnitude and final-model prediction.

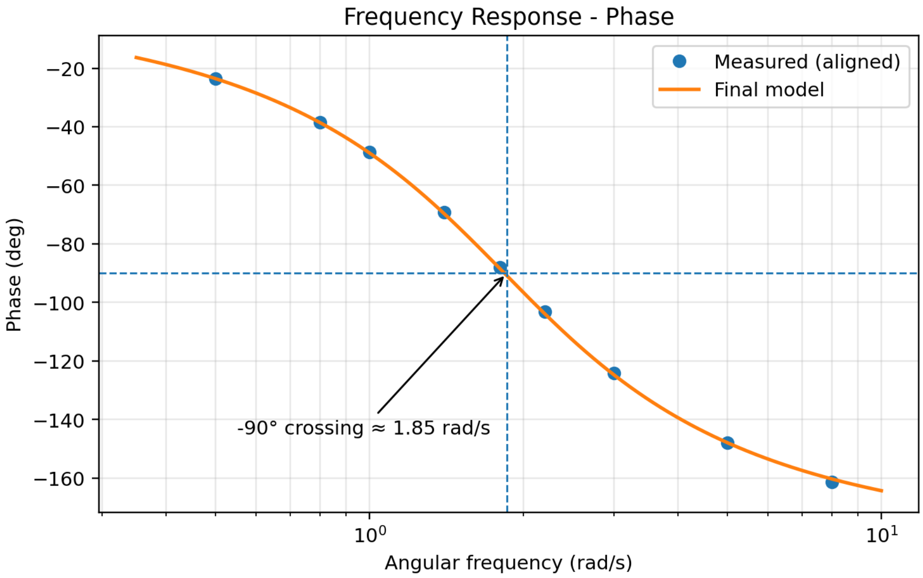

Figure 4: Aligned measured phase and final-model prediction. The $-90°$ crossing occurs close to the natural frequency.

For a standard second-order low-pass model, the phase is exactly $-90°$ at

$$\omega = \omega_n \tag{32}$$

Linear interpolation between the measured points at

$$1.8 \text{ rad/s} \quad \text{and} \quad 2.2 \text{ rad/s}$$

gives

$$\omega_{n,\text{freq}} \approx \mathbf{1.854}\,\text{rad/s} \tag{33}$$

The lowest measured frequency gives the low-frequency gain estimate

$$K_{\text{freq}} \approx |G(j0.5)| = \mathbf{0.631} \tag{34}$$

At $\omega = \omega_n$, a second-order model satisfies

$$|G(j\omega_n)| = \frac{K}{2\zeta} \tag{35}$$

Interpolating the measured magnitude at the $-90°$ crossing gives

$$|G(j\omega_n)| \approx \mathbf{0.429} \tag{36}$$

Hence,

$$\zeta_{\text{freq}} = \frac{K}{2|G(j\omega_n)|} = \frac{0.631}{2(0.429)} = \mathbf{0.735} \tag{37}$$

The independently identified frequency-domain model is therefore

$$\boxed{G_{\text{freq}}(s) = \frac{2.170}{s^2 + 2.726s + 3.437}} \tag{38}$$

with poles

$$s = \mathbf{-1.363 \pm j1.257} \tag{39}$$

These results closely confirm the gain, damping ratio, natural frequency and pole locations obtained from the step response.

3.3 Proposed System Model

Table 6 compares the independent parameter estimates.

The natural-frequency estimates differ by approximately 2.3%, while the damping-ratio estimates differ by approximately 1.4% relative to their average.

A final model was therefore formed by averaging

$$K, \qquad \zeta, \qquad \omega_n$$

from the two experiments.

Table 6: Comparison of identified parameters.

Estimate	K	ζ	ω_n **(rad/s)**
Step test	0.638	0.745	1.813
Frequency test	0.631	0.735	1.854
Final average	**0.635**	**0.740**	**1.833**

The proposed model is

$$\boxed{G(s) = \frac{X(s)}{U(s)} = \frac{2.133}{s^2 + 2.715s + 3.361}} \tag{40}$$

with poles

$$\boxed{s = -\mathbf{1.357 \pm j1.233}} \tag{41}$$

Using Equation (6), the final spring and damping estimates for the normalized $m = 1$ system are

$$\boxed{\boldsymbol{b \approx 2.715}}, \qquad \boxed{\boldsymbol{k \approx 3.361}} \tag{42}$$

The numerator

$$\boldsymbol{c \approx 2.133} \tag{43}$$

represents the proportionality between the simulator input and the applied-force term in the normalized model.

The final model parameters are collected in Table 7.

Table 7: Final proposed second-order model parameters.

Quantity	**Symbol**	**Final estimate**
DC gain	K	**0.635**
Damping ratio	ζ	**0.740**
Natural frequency	ω_n	**1.833** rad/s
Damping coefficient	b	**2.715**
Spring coefficient	k	**3.361**
Input-force proportionality	c	**2.133**
Poles	$s_{1,2}$	$-\mathbf{1.357 \pm j1.233}$

3.4 Integral Action for Zero Steady-State Error

The proposed model contains no integrator, so proportional control alone would generally leave a non-zero steady-state error for a constant reference.

Adding integral action,

$$C(s) = \frac{K_i}{s}, \tag{44}$$

changes the open-loop system to Type 1. For unity feedback,

$$L(s) = \frac{2.133K_i}{s\,(s^2 + 2.715s + 3.361)}, \tag{45}$$

and therefore, for a step reference,

$$e_{ss} = \boxed{\mathbf{0}}, \tag{46}$$

provided the closed-loop system is stable.

The characteristic equation becomes

$$s^3 + 2.715s^2 + 3.361s + 2.133K_i = 0. \tag{47}$$

Integral action therefore removes the steady-state step-tracking error, although K_i must still be tuned to obtain acceptable transient performance. Controller design was outside the scope of this experiment.

3.5 Model Validation

The final model was tested against both experiments.

Figure 5 compares the measured zero-adjusted step response to the response predicted by Equation (40) for the same 4-unit input.

The model reproduces the rise, small overshoot and settling behaviour closely.

Over the first 12 seconds after the step, the root-mean-square displacement error is approximately

$$\mathbf{0.028}.$$

This corresponds to approximately

$$\mathbf{1.1\%}$$

of the measured steady-state output change.

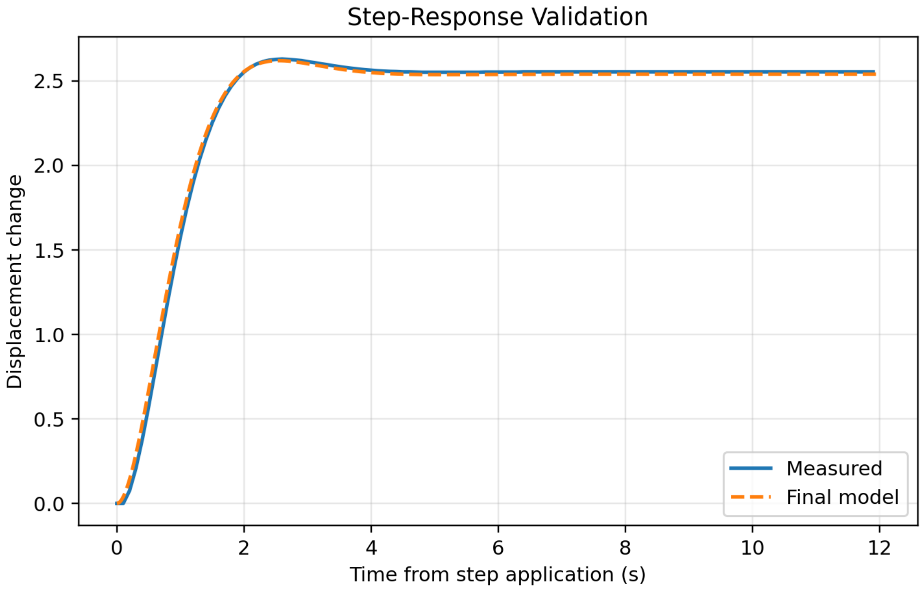

Figure 5: Validation of the proposed model against the measured step response.

The same model also tracks the measured Bode data in Figures 3 and 4.

Across all nine tested frequencies, the RMS magnitude error is approximately

$$\mathbf{0.27}\ \mathrm{dB},$$

and the RMS phase error after sample-alignment correction is approximately

$$\mathbf{0.51}^\circ.$$

The largest discrepancies remain below approximately

$$\mathbf{0.65}\ \mathrm{dB}$$

in magnitude and

$$\mathbf{1}^\circ$$

in phase.

The strong agreement in both domains indicates that a second-order model is sufficient to describe the dominant suspension dynamics over the measured range.

The validation statistics are summarised in Table 8.

Table 8: Summary of model-validation errors.

Validation quantity	**Error measure**	**Result**
Step response	RMS displacement error	**0.028**
Step response	RMS error relative to steady-state change	**1.1%**
Bode magnitude	RMS error	**0.27** dB
Bode magnitude	Maximum error	$<$ **0.65** dB
Bode phase	RMS error	**0.51** deg
Bode phase	Maximum error	$<$ **1** deg

4 Conclusion

A second-order model of the suspension simulator was successfully identified from independent step- and frequency-response tests.

The independent estimates are compared in Table 6, and the final parameter set is summarised in Table 7.

The step response had a steady-state gain of approximately **0.638**, a small overshoot of **2.98%**, and a peak time of **2.6 s**, giving

$$\zeta \approx \mathbf{0.745},$$

$$\omega_n \approx \mathbf{1.813}\,\mathrm{rad/s},$$

and poles at approximately

$$\mathbf{-1.351 \pm j1.208}.$$

The frequency test independently produced

$$K \approx \mathbf{0.631},$$

$$\zeta \approx \mathbf{0.735},$$

and

$$\omega_n \approx \mathbf{1.854}\,\mathrm{rad/s},$$

confirming the time-domain result.

Averaging the two parameter sets gave the proposed transfer function

$$\boxed{G(s) = \frac{2.133}{s^2 + 2.715s + 3.361}}.$$

This corresponds to

$$\boldsymbol{b \approx 2.715}$$

and

$$\boldsymbol{k \approx 3.361}$$

for the normalized $m = 1$ system.

The model closely reproduced both the measured step response and all nine frequency-response points.

The absence of a resonant peak in the Bode magnitude is also consistent with the estimated damping ratio being slightly greater than

$$\frac{1}{\sqrt{2}}.$$

Overall, the identified transfer function provides a simple and well-validated representation of the dominant dynamics of the experimental system.

A Appendix and Notes

The complete analysis has been archived in a public GitHub repository.

Instead of reproducing several hundred lines of MATLAB code and large raw CSV datasets inside this report, the appendix provides direct links to the exact files used to generate the results.

This keeps the report concise while preserving full reproducibility.

Open the complete EEE3094S Lab 1 GitHub repository

A.1 MATLAB Analysis Code

The two MATLAB scripts used for identification and validation are available directly from the repository.

Table 9: Clickable links to MATLAB analysis scripts.

File	Purpose
Step-response MATLAB script	Reads the raw step-test CSV, identifies the step, calculates gain, overshoot, damping ratio, natural frequency and poles, constructs the step-test transfer function, validates the response and exports step-response figures.
Bode/frequency MATLAB script	Reads the nine frequency-test CSV files, estimates sinusoidal amplitudes and phase, constructs the measured Bode data, estimates the frequency-domain model, compares it with the step model and exports the Bode-validation figures.

A.2 Experimental and Processed Data

The raw laboratory files and MATLAB-generated result tables are available through the links in Table 10.

Table 10: Clickable links to experimental and processed data.

Resource	Contents
Raw step-test data	Original step-response CSV together with the laboratory step-test images.
Raw frequency-test data	Original CSV files for the nine tested frequencies: 0.5, 0.8, 1.0, 1.4, 1.8, 2.2, 3.0, 5.0 and 8.0 rad/s.
Processed step-identification results	MATLAB-generated summary of the identified time-domain parameters.
Processed frequency-response results	MATLAB-generated magnitude and phase results for the sinusoidal tests.
Model comparison results	MATLAB-generated comparison of the step-test, frequency-test and final combined models.

A.3 Figures and Report Source

For completeness, the generated figures and report files are also available online.

Table 11: Clickable links to generated figures and report files.

Resource	Description
Figures folder	Exported step-response, Bode and model-validation figures used in this report.
LaTeX source (`main.tex`)	Source used to compile the report, including clickable contents, figure/table references and repository links.
Final PDF report	Compiled version of the submitted Lab 1 report.
Repository home page	Complete project structure, README and version history.

A.4 Reproducibility Note

To reproduce the analysis, download or clone the repository and keep the

`Data`, `Matlab`, `Figures`

folders in their current relative locations.

Run

`step_response_validation.m`

followed by

`bode_validation.m`

in MATLAB.

The scripts read the experimental CSV files and regenerate the identification results and figures used in this report.

References

[1] EEE3094S Control Systems Engineering, *Lab 1: System Identification*, University of Cape Town, 2026.

[2] UCT Control Laboratory, *Control Laboratory Suspension Simulator User Manual*, Release 1.0, 10 Aug. 2023.

[3] J. Pieters, "EEE3094S Lab 1 – System Identification: Experimental Data, MATLAB Analysis and Report Source," GitHub repository, 2026. [Online]. Available: https://github.com/NateDogg247/EEE3094S---Lab1-System-Identification